

WELCOME TO

Casa Azul Supply Co.

Sales Data & SQL Analysis Booklet

A beginner-friendly guide to the data and ten SQL questions

Furniture · Office Supplies · Technology · 2014 – 2017

Welcome to Casa Azul Supply Co.

Casa Azul means “Blue House” in Spanish, and the blue tiles on every page are our logo. Casa Azul Supply Co. sells **furniture**, **office supplies** and **technology** across the United States. This booklet walks through four years of its sales data and answers ten classic questions with SQL, step by step.

$2,297,201 Total sales	**$286,397** Total profit	**9,994** Rows of data
5,009 Orders placed	**793** Customers	**1,850** Different products

The data covers **January 2014 – December 2017**, with 3 categories, 17 sub-categories, 4 regions and 49 states. Each row is one product line inside an order, so a single order can span several rows.

How to read each question

1. **The question**
   A plain-English business question, such as “Who are our best customers?”
2. **The SQL**
   The query we wrote in SQL Server. Blue words are SQL keywords.
3. **The result**
   A chart or table of what the query returned, plus what we learned.

The ten questions

Q1 Top customers	Q6 Loss-making products
Q2 Latest orders	Q7 Biggest orders
Q3 Best-selling products	Q8 Sales per category
Q4 Product categories	Q9 Sales regions
Q5 Orders over $1,000	Q10 Units sold per product

Meet the data: 21 columns

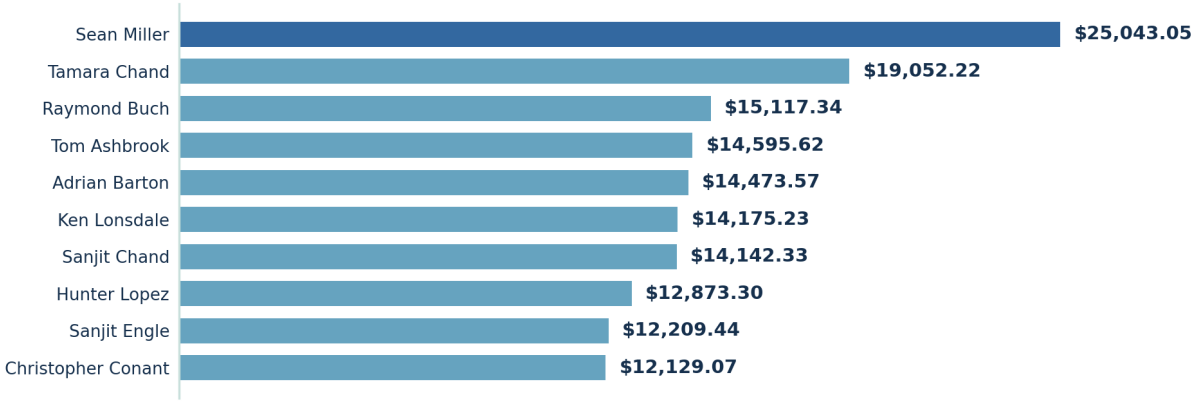
Every column belongs to one of five families. In SQL Server the headers use underscores — for example **Order ID** becomes **Order_ID** — and the table is named **dbo.Supersales**.

#	Column	Family	What it tells us
1	**Row ID**	Orders	A unique number for each row
2	**Order ID**	Orders	Code of the order, e.g. CA-2016-152156
3	**Order Date**	Orders	Day the order was placed
4	**Ship Date**	Orders	Day the order was shipped
5	**Ship Mode**	Orders	Shipping speed: Standard, Second, First Class or Same Day
6	**Customer ID**	Customers	Unique code of the customer
7	**Customer Name**	Customers	Full name of the customer
8	**Segment**	Customers	Consumer, Corporate or Home Office
9	**Country**	Geography	Country (United States)
10	**City**	Geography	Customer's city
11	**State**	Geography	US state (49 appear in the data)
12	**Postal Code**	Geography	ZIP code
13	**Region**	Geography	Central, East, South or West
14	**Product ID**	Products	Unique code of the product
15	**Category**	Products	Furniture, Office Supplies or Technology
16	**Sub-Category**	Products	17 finer groups, e.g. Chairs or Binders
17	**Product Name**	Products	Full name of the product
18	**Sales**	Numbers	Money earned on the row, in US dollars
19	**Quantity**	Numbers	Number of units sold
20	**Discount**	Numbers	Discount given (0 = none, 0.2 = 20%)
21	**Profit**	Numbers	Profit or loss on the row, in US dollars

Q1 Who are our top customers?

We add up every purchase each customer made, then keep the 10 who spent the most. **GROUP BY** folds many rows into one row per customer, and **DESC** puts the biggest total first.

```
SELECT TOP 10 Customer_Name, SUM(Sales) AS Total_Sales
FROM dbo.Supersales
GROUP BY Customer_Name
ORDER BY Total_Sales DESC;
```



WHAT WE LEARNED

Sean Miller leads with $25,043, followed by Tamara Chand at $19,052.

Q2 What were the latest orders?

We sort all rows by date, newest first, and keep the first 10. **TOP 10** trims the list after sorting.

```
SELECT TOP 10 Order_ID, Order_Date, Customer_Name, Sales
FROM dbo.Supersales
ORDER BY Order_Date DESC;
```

Order ID	Order date	Customer	Sales
CA-2017-126221	2017-12-30	Chuck Clark	$209.30
CA-2017-143259	2017-12-30	Patrick O'Donnell	$323.14
CA-2017-143259	2017-12-30	Patrick O'Donnell	$90.93
CA-2017-143259	2017-12-30	Patrick O'Donnell	$52.78
CA-2017-115427	2017-12-30	Erica Bern	$13.90
CA-2017-115427	2017-12-30	Erica Bern	$20.72
CA-2017-156720	2017-12-30	Jill Matthias	$3.02
CA-2017-118885	2017-12-29	James Galang	$393.57
CA-2017-118885	2017-12-29	James Galang	$302.38
US-2017-102638	2017-12-29	Michael Chen	$6.03

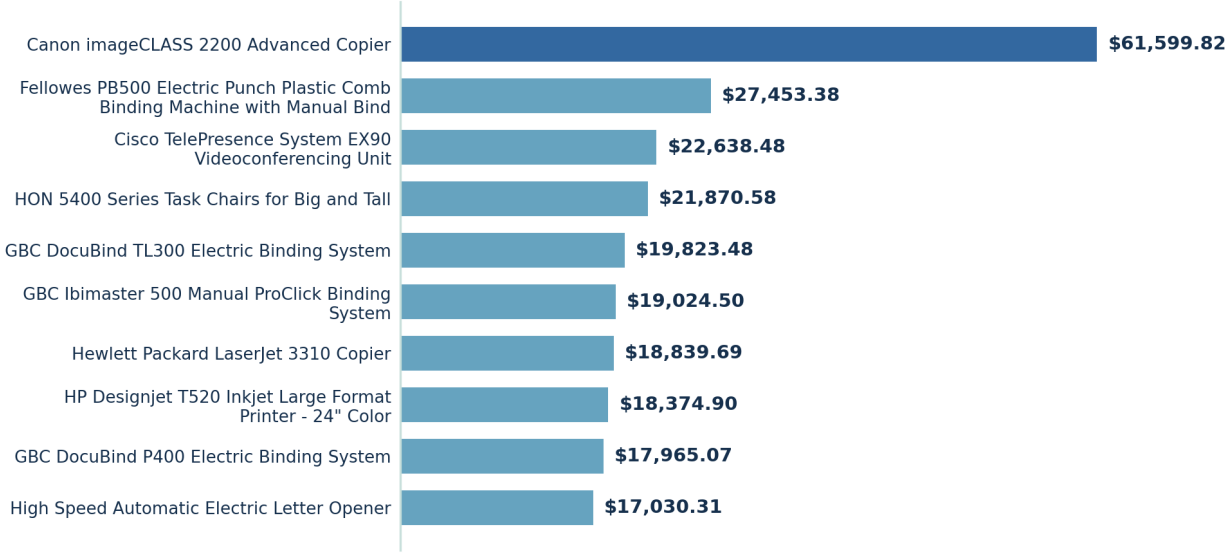
WHAT WE LEARNED

All ten rows fall on 29–30 December 2017, the very end of our data. Order CA-2017-143259 shows up three times because it holds three products.

Q3 Which products sell best?

Same idea as Q1, but we group by **product** instead of customer and add up sales revenue.

```
SELECT TOP 10 Product_Name, SUM(Sales) AS Total_Sales
FROM dbo.Supersales
GROUP BY Product_Name
ORDER BY Total_Sales DESC;
```



WHAT WE LEARNED

The Canon imageCLASS 2200 copier earns $61,600 — more than double the runner-up. Copiers, binding machines and chairs fill the list.

Q4 Which product categories exist?

DISTINCT removes duplicates, so each category is listed exactly once.

```
SELECT DISTINCT Category
FROM dbo.Supersales
ORDER BY Category;
```

Furniture	Office Supplies	Technology

WHAT WE LEARNED

The store sells three product categories. We will see how much each one earns in Q8.

Q5 Which orders top $1,000?

WHERE keeps only the rows that pass a test. Here, only order lines with sales above 1,000 stay in the result.

```
SELECT Order_ID, Product_Name, Sales
FROM dbo.Supersales
WHERE Sales > 1000
ORDER BY Sales DESC;
```

Order ID	Product (first 6 of 468 rows)	Sales
CA-2014-145317	Cisco TelePresence System EX90 Videoconferencing Unit	$22,638.48
CA-2016-118689	Canon imageCLASS 2200 Advanced Copier	$17,499.95
CA-2017-140151	Canon imageCLASS 2200 Advanced Copier	$13,999.96
CA-2017-127180	Canon imageCLASS 2200 Advanced Copier	$11,199.97
CA-2017-166709	Canon imageCLASS 2200 Advanced Copier	$10,499.97
CA-2016-117121	GBC Ibimaster 500 Manual ProClick Binding System	$9,892.74

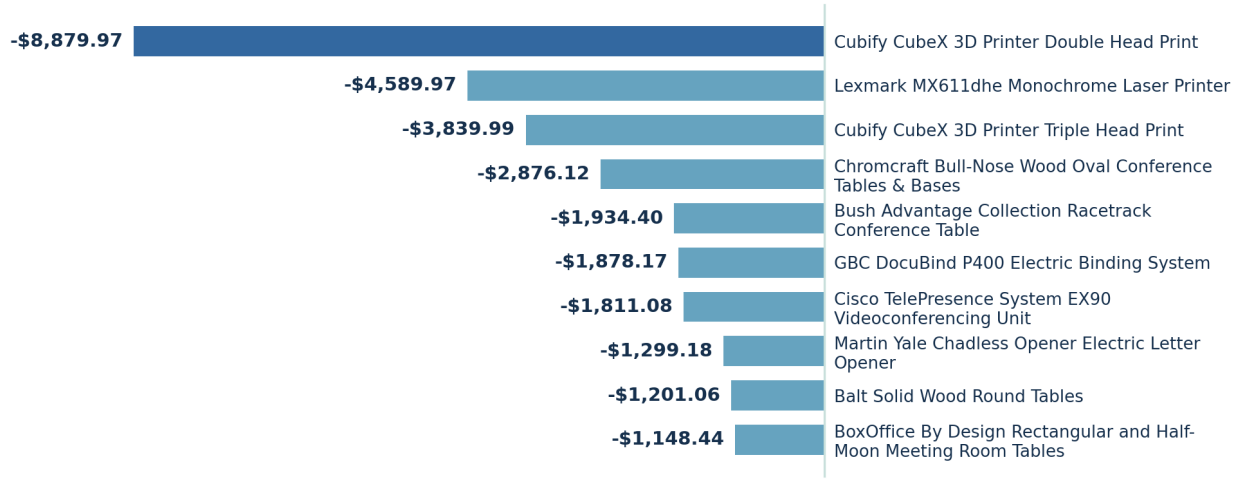
WHAT WE LEARNED

468 order lines pass $1,000: 177 Technology, 169 Furniture and 122 Office Supplies. Big tickets come from machines and furniture.

Q6 Which products lose money?

HAVING filters *after* grouping (WHERE filters before). It keeps only products whose total profit is below zero.

```
SELECT Product_Name, SUM(Profit) AS Total_Profit
FROM dbo.Supersales
GROUP BY Product_Name
HAVING SUM(Profit) < 0
ORDER BY Total_Profit ASC;
```



WHAT WE LEARNED

299 products lost money overall (chart shows the ten biggest). The Cubify CubeX 3D printers are the biggest drain:

Q7 The ten biggest orders

Like Q5, but with no filter: we sort every row by sales and keep the first ten.

```
SELECT TOP 10 Order_ID, Order_Date, Product_Name, Sales
FROM dbo.Supersales
ORDER BY Sales DESC;
```

Order ID	Date	Product	Sales
CA-2014-145317	2014-03-18	Cisco TelePresence System EX90 Videoconferencing Unit	$22,638.48
CA-2016-118689	2016-10-02	Canon imageCLASS 2200 Advanced Copier	$17,499.95
CA-2017-140151	2017-03-23	Canon imageCLASS 2200 Advanced Copier	$13,999.96
CA-2017-127180	2017-10-22	Canon imageCLASS 2200 Advanced Copier	$11,199.97
CA-2017-166709	2017-11-17	Canon imageCLASS 2200 Advanced Copier	$10,499.97
CA-2016-117121	2016-12-17	GBC Ibimaster 500 Manual ProClick Binding System	$9,892.74
CA-2014-116904	2014-09-23	Ibico EPK-21 Electric Binding System	$9,449.95
US-2016-107440	2016-04-16	3D Systems Cube Printer, 2nd Generation, Magenta	$9,099.93
CA-2016-158841	2016-02-02	HP Designjet T520 Inkjet Large Format Printer - 24" Color	$8,749.95
CA-2016-143714	2016-05-23	Canon imageCLASS 2200 Advanced Copier	$8,399.98

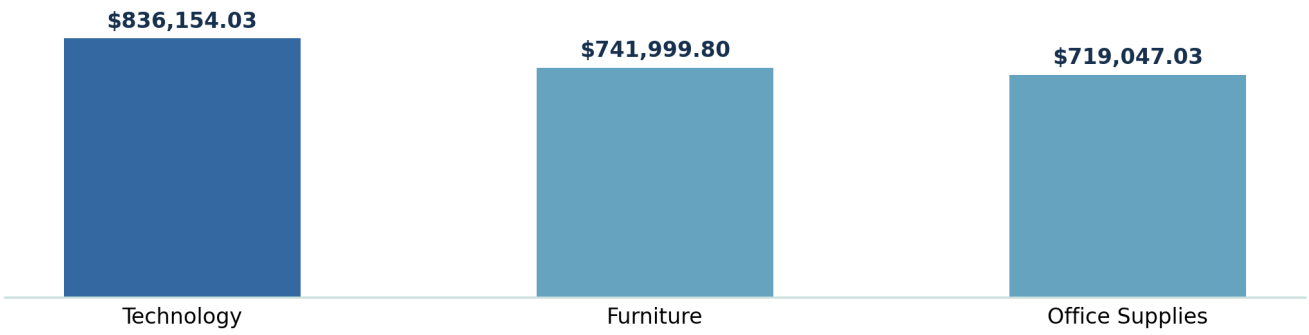
WHAT WE LEARNED

The largest single line is $22,638 (Cisco TelePresence, March 2014). Canon copiers take five of the ten places.

Q8 Sales by product category

A classic summary: one total per category, largest first.

```
SELECT Category, SUM(Sales) AS Total_Sales
FROM dbo.Supersales
GROUP BY Category
ORDER BY Total_Sales DESC;
```



WHAT WE LEARNED

Technology leads with $836,154.03 (36%), then Furniture (32%) and Office Supplies (31%). The three categories are remarkably balanced.

Q9 Which regions are in the data?

Another **DISTINCT** query: the list of sales regions, each shown once.

```
SELECT DISTINCT Region
FROM dbo.Supersales
ORDER BY Region;
```

Central	East	South	West

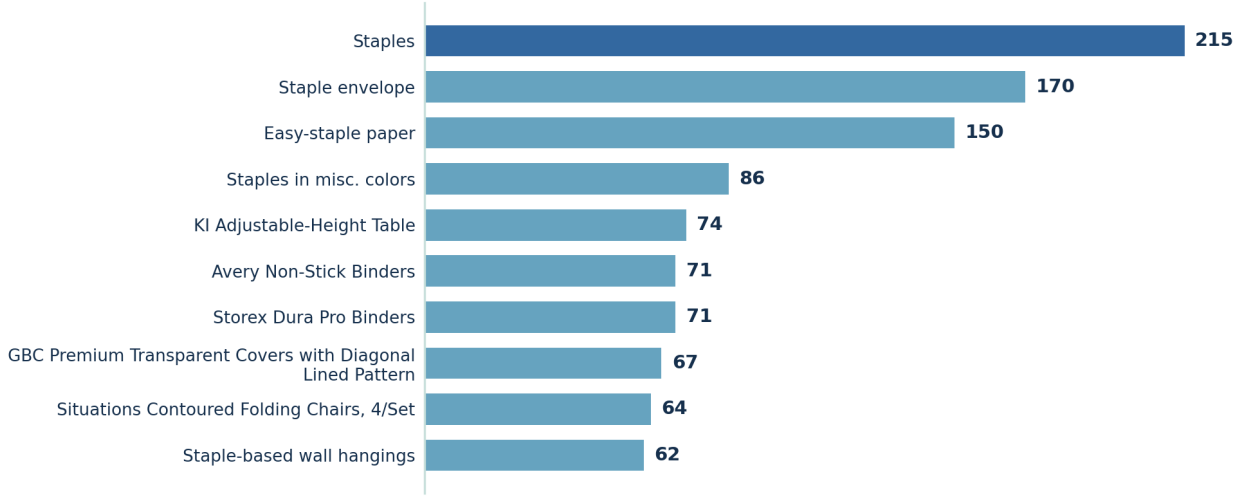
WHAT WE LEARNED

Sales are split across four US regions. The next page shows how much each one sells.

Q10 Units sold per product

This time we add up **Quantity** (units) instead of Sales (dollars).

```
SELECT Product_Name, SUM(Quantity) AS Total_Quantity_Sold
FROM dbo.Supersales
GROUP BY Product_Name
ORDER BY Total_Quantity_Sold DESC;
```



WHAT WE LEARNED

Staples-related products take the first four places, led by plain Staples at 215 units. Cheap items win on volume, copiers win on value (Q3).

What the data told us

$836K
Technology is our top category by sales, just ahead of Furniture and Office Supplies.
SOURCE: Q8

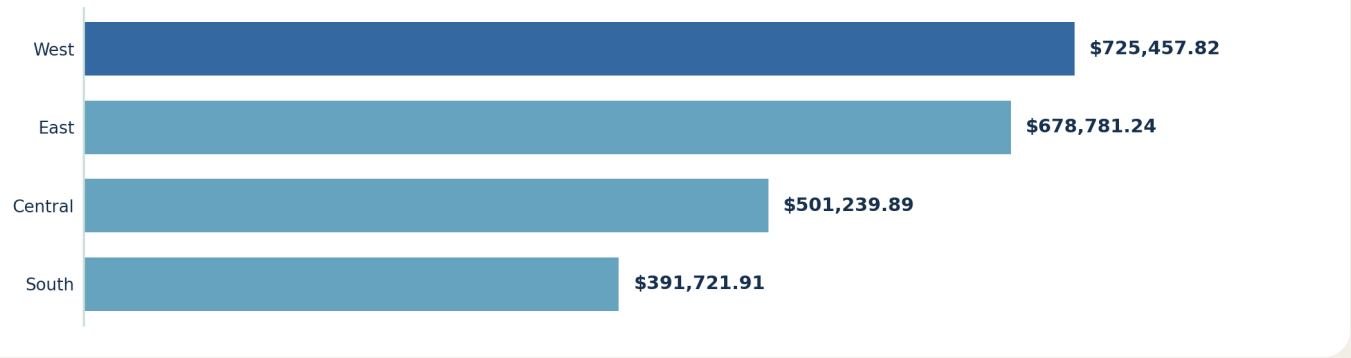
$61.6K
One product, the Canon imageCLASS 2200 copier, out-earns every other item in the store.
SOURCE: Q3

299
products finished with a net loss — 3D printers and conference tables lead the list.
SOURCE: Q6

215
units of Staples sold: the most popular product by volume, a humble best-seller.
SOURCE: Q10

Bonus: sales by region

Q9 told us which regions exist. Adding up sales for each one shows the West and East lead the way.



Where to go next

- Group sales by year to see how the store grew from 2014 to 2017.
- Compare profit per region, not only sales.
- Check how discounts relate to losses, starting with the products from Q6.
- Join this table with a targets table to track goals.

Key insights

Five patterns that matter most. They come from the same 9,994 rows: sales, profit and discount.

+51%

Sales grew every year after 2015

Sales rose from $484K (2014) to $733K (2017). Every year peaks in September, November and December.

SOURCE: Sales by year and month

2.5%

Furniture earns almost no profit

Technology earns 17.4% and Office Supplies 17.0%, yet all three categories sell about the same (Q8).

SOURCE: Profit margin by category

–10%

Deep discounts turn sales into losses

Margin is –10.0% at 21–30% discount, versus +29.5% with no discount. Lines discounted 21% or more lost about $135K in total.

SOURCE: Margin by discount level

–$1,981

Big sales are not always big profit

Sean Miller is our #1 customer by sales (Q1) but lost us $1,981. The #3 best-seller, Cisco TelePresence (Q3), also lost money. 299 of 1,850 products (16.2%) lose money (Q6).

SOURCE: Q1, Q3, Q6 + profit data

7.9%

Central has the weakest margin

Central gives the highest average discount (24.0% vs 10.9% in West). Texas (–$25.7K) and Illinois (–$12.6K) lose the most there.

SOURCE: Profit and discount by region

Read with care: these are patterns in the data, not proof of cause. Discounts and losses move together, but more analysis is needed to confirm why.

Recommendations

Each action follows the evidence above. Start with the three that stop money from leaking.

ACT FIRST · STOP THE PROFIT LEAKS

1 Review discounts above 20%

EVIDENCE
Lines discounted 21% or more lose money (about –$135K).

ACTION
Set a discount ceiling or an approval step, starting with Tables and Bookcases.

HOW TO MEASURE (proposed)
Share of sales lines above 20% discount; margin of discounted lines.

2 Fix Furniture pricing

EVIDENCE
Furniture margin is 2.5%; Tables –$17.7K, Bookcases –$3.5K.

ACTION
Investigate costs and discounts on Tables and Bookcases before changing prices or range.

HOW TO MEASURE (proposed)
Margin by sub-category after the review.

3 Review Central region discounts

EVIDENCE
Central: 24.0% average discount, 7.9% margin.

ACTION
Audit discount practice in Texas and Illinois, then check Ohio and Pennsylvania (East).

HOW TO MEASURE (proposed)
Average discount and margin in those states.

BUILD NEXT · GROW WHERE IT PAYS

4 Prepare for September–December

EVIDENCE
September, November and December are the strongest months every year.

ACTION
Plan campaigns and stock ahead of the autumn peak; study why February is weak.

HOW TO MEASURE (proposed)
Sep–Dec sales vs prior year; February sales.

5 Reward the 88 loyal customers

EVIDENCE
88 customers (11.1%) with 10+ orders bring 19.3% of sales.

ACTION
Create a loyalty programme; check low-profit loyal accounts first.

HOW TO MEASURE (proposed)
Orders and profit per loyal customer.

6 Track profit next to sales

EVIDENCE
Sean Miller is #1 in sales but lost $1,981; 155 customers have negative profit.

ACTION
Show profit beside sales in customer and product rankings; investigate the 299 loss-making products.

HOW TO MEASURE (proposed)
Number of loss-making products and customers.

Success measures are proposed and not yet calculated. No financial impact is claimed.

SQL cheat sheet

Every keyword used in this booklet, in plain words.

Keyword	What it does	Used in
`SELECT`	Choose which columns to show	All
`TOP n`	Keep only the first n rows	Q1, Q2, Q3, Q7
`DISTINCT`	Show each value only once	Q4, Q9
`FROM`	Name the table to read	All
`WHERE`	Filter rows before grouping	Q5
`GROUP BY`	Make one row per group	Q1, Q3, Q6, Q8, Q10
`HAVING`	Filter groups after grouping	Q6
`ORDER BY`	Sort the result (ASC or DESC)	All
`SUM( )`	Add up the values	Q1, Q3, Q6, Q8, Q10

THE END

Thank you

That’s the end of our tour through Casa Azul Supply Co.’s sales story. Questions are always welcome.